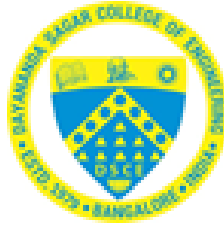


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Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru-111



Report

On

Interdisciplinary Project Based Learning - 1BPRJ208

“RIVER FLOOD DETECTION SYSTEM”

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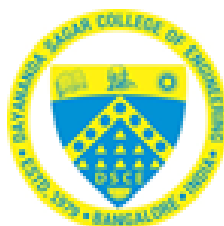
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CERTIFICATE

This is to certify that the project entitled “**RIVER FLOOD DETECTION SYSTEM**” is a bonafide work carried out by **PURURJITH KADEKAR [1DS25ET087]**, **PRANAV VERMA [1DS25ET084]**, **ROSHANI KUMARI [1DS25CS212]**, **RUPESH KUMAR S [1DS25CS213]** in partial fulfilment of 2nd semester, Bachelor of Engineering in Computer Science and Engineering and Electronics and Telecommunication Engineering under Visvesvaraya Technological University, Belgaum during the year 2025-26.

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CONTENTS

SL. NO	CONTENT	PG. NO
1.	Abstract	3
2.	Introduction	4
3.	Data Collection	6
4.	System design & Methodology	7
5.	Snapshots and Results	16
6.	Conclusion & Future Enhancements	22
7.	References	24

ABSTRACT

Floods are among the most destructive natural disasters, causing significant loss of life, property damage, and environmental disruption. Timely detection and early warning systems are essential for minimizing these impacts, especially in remote areas where conventional monitoring infrastructure is limited or unavailable. The proposed **River Flood Detection System** is an intelligent, cost-effective, and sustainable solution designed to provide continuous monitoring of river water levels and issue timely flood warnings.

The system utilizes an **ESP32-S3 microcontroller** integrated with an **HC-SR04 ultrasonic sensor** to measure water levels in real time. The sensor continuously monitors the distance between the water surface and the sensing unit, enabling accurate detection of rising water levels. When the measured level exceeds a predefined threshold, a local buzzer is activated to provide an immediate warning to nearby residents and authorities. To ensure reliable communication over long distances, the system employs **LoRa (Long Range) wireless technology**, which transmits sensor data to a central monitoring station without relying on cellular networks.

A **Raspberry Pi Zero 2W** serves as the gateway, receiving and processing sensor readings. The collected data is stored in a local **SQLite database**, ensuring uninterrupted logging even during internet outages. The gateway also hosts a web-based dashboard developed using **Flask**, allowing users to monitor river conditions remotely through real-time graphs and status updates. To prevent false alarms caused by temporary disturbances, a temporal verification algorithm analyzes multiple sensor readings before confirming a flood event.

For emergency communication, the system uses a smartphone-based SMS gateway through the **TextBee API**, enabling instant text message alerts to designated users without requiring expensive GSM hardware. The entire sensing node is powered by a **solar energy system** consisting of a solar panel, charge controller, and rechargeable lithium-ion battery, ensuring continuous operation in off-grid locations.

The proposed River Flood Detection System combines IoT, wireless communication, renewable energy, and real-time data analytics to create a dependable flood monitoring platform. Its low cost, energy efficiency, and infrastructure-independent operation make it highly suitable for deployment in rural and flood-prone regions, helping communities improve disaster preparedness and reduce flood-related risks.

Chapter 1

INTRODUCTION

Floods are among the most destructive natural disasters affecting millions of people worldwide every year. They cause significant loss of human life, damage to infrastructure, destruction of agricultural lands, and severe economic setbacks. In many regions, especially rural and remote areas, floods occur with little warning, leaving communities vulnerable and unprepared. Climate change, irregular rainfall patterns, rapid urbanization, and deforestation have further increased the frequency and intensity of flooding events, making effective flood monitoring and early warning systems more important than ever.

Traditional flood monitoring methods often depend on manual observation, periodic inspections, or expensive monitoring infrastructure. These approaches are generally inefficient because they do not provide continuous real-time information about changing river conditions. In many remote locations, access to reliable electrical power and communication networks is limited, making the deployment of conventional monitoring systems difficult and costly. As a result, there is a growing need for low-cost, autonomous, and reliable flood detection systems that can operate independently and provide timely alerts to authorities and local residents.

Recent advancements in the Internet of Things (IoT), wireless communication technologies, and embedded systems have enabled the development of smart environmental monitoring solutions. IoT-based monitoring systems utilize sensors, microcontrollers, and communication modules to collect real-time data from the environment and transmit it to users for analysis and decision-making. Such systems offer improved accuracy, reduced operational costs, and the ability to provide instant warnings during emergency situations.

The River Flood Detection System is designed to address these challenges by implementing an intelligent and cost-effective flood monitoring solution. The system continuously monitors river water levels using an ultrasonic sensor that measures the distance between the sensor and the water surface. The collected data is processed by an ESP32-S3 microcontroller, which serves as the primary sensing and control unit. By continuously analyzing water-level variations, the system can identify potentially dangerous conditions and initiate warning procedures before severe flooding occurs.

A key feature of the proposed system is the use of LoRa (Long Range) wireless communication technology. LoRa enables data transmission over several kilometers while consuming very little power. Unlike conventional communication systems that rely on cellular networks or internet connectivity, LoRa provides reliable long-range communication even in isolated locations. This makes it highly suitable for flood monitoring applications where sensor nodes may be installed near rivers, dams, or remote water bodies.

To ensure uninterrupted operation, the system is powered by a renewable energy source consisting of a solar panel, rechargeable lithium-ion battery, and charging controller circuit. The solar-powered design allows the monitoring station to function continuously without requiring frequent battery replacement or external power connections. This feature significantly improves system reliability and makes it suitable for long-term deployment in remote environments.

The collected sensor data is transmitted through the LoRa network to a Raspberry Pi Zero 2W gateway. The gateway acts as the central processing unit of the system and performs several important functions, including data storage, flood verification, visualization, and alert generation. Water-level measurements are stored in a local SQLite database, ensuring that historical records remain available even during internet outages. The gateway also hosts a web-based dashboard that allows users to monitor real-time river conditions from any connected device.

RIVER FLOOD DETECTION SYSTEM

To improve the reliability of flood warnings, the system incorporates a temporal verification algorithm. Instead of generating alerts based on a single sensor reading, multiple consecutive readings are analyzed over a specified time interval. This approach helps eliminate false alarms caused by temporary disturbances such as floating debris, heavy rainfall splashes, or sensor noise. Only when sustained abnormal water levels are detected does the system classify the event as a confirmed flood condition.

Another innovative aspect of the project is its low-cost emergency notification mechanism. Instead of using dedicated GSM modules, which can increase system cost and complexity, the proposed solution utilizes a repurposed smartphone running an SMS gateway application. When a flood condition is confirmed, the Raspberry Pi sends a command through a REST API to the smartphone, which then dispatches SMS alerts using its native cellular messaging service. This approach reduces hardware expenses while maintaining effective communication during emergencies.

The proposed River Flood Detection System combines modern sensing technology, long-range wireless communication, renewable energy, local data storage, and automated alert mechanisms into a single integrated platform. The system is designed to provide accurate, real-time flood monitoring while remaining affordable, scalable, and easy to deploy. By offering early warnings and continuous monitoring capabilities, the project contributes to disaster preparedness and helps protect communities, infrastructure, and valuable resources from the devastating effects of floods.

Ultimately, the development of this system demonstrates how emerging technologies can be applied to solve real-world environmental challenges. The project promotes sustainable engineering practices through the use of solar energy and reusable hardware while providing a practical solution for enhancing public safety and reducing flood-related risks in vulnerable regions.

Chapter 2

DATA COLLECTION

The data collection process for the River Flood Detection System is designed to continuously monitor river water levels and provide reliable information for flood prediction and alert generation. The system collects real-time environmental data using an HC-SR04 ultrasonic sensor mounted above the river surface. This sensor measures the distance between the sensor and the water level by transmitting ultrasonic waves and calculating the time taken for the reflected signal to return.

The sensing unit is built around the ESP32-S3 microcontroller, which acquires distance measurements at regular intervals. The collected readings are processed locally and converted into water-level values. To ensure accuracy and consistency, measurements are taken every five seconds and filtered to eliminate noise caused by rain, floating debris, or temporary disturbances on the water surface.

Once the data is collected, it is transmitted wirelessly through the LoRa (Long Range) communication module operating at 433 MHz. LoRa technology enables reliable communication over several kilometers, making it suitable for remote riverbank locations where conventional cellular connectivity may be unavailable. Each transmitted packet contains the measured water level, timestamp, and communication quality parameters such as RSSI (Received Signal Strength Indicator).

At the gateway side, a Raspberry Pi Zero 2W receives the transmitted data and stores it in a local SQLite database. The database maintains a continuous historical record of water-level measurements, allowing trend analysis and future reference. This local storage mechanism ensures that data remains available even during internet outages, improving the reliability of the monitoring system.

To verify flood conditions, the collected data is analyzed using a temporal verification algorithm. A rolling buffer of recent measurements is maintained, and flood alerts are generated only when a significant percentage of readings exceed the predefined threshold. This method reduces false alarms and improves the accuracy of flood detection.

The stored data is also displayed on a web-based dashboard developed using Flask and Chart.js. The dashboard provides real-time visualization of water-level trends, enabling authorities and local communities to monitor river conditions remotely. During critical situations, the collected data triggers SMS notifications through the TextBee API, ensuring timely dissemination of flood warnings.

Thus, the data collection system provides continuous, accurate, and reliable monitoring of river water levels, forming the foundation for effective flood detection and early warning operations.

Chapter 3

SYSTEM DESIGN & METHODOLOGY

3.1 System Design

3.1.1 Introduction

The River Flood Detection System is designed to provide an early warning mechanism for detecting rising river water levels and alerting nearby communities before a flood situation occurs. The system combines IoT technology, wireless communication, renewable energy sources, and real-time monitoring to create a reliable flood detection network suitable for remote and rural environments.

The proposed system consists of two major sections:

1. **Edge Monitoring Node**
2. **Central Gateway and Alert System**

The edge node is installed near the riverbank and continuously measures the distance between the sensor and the water surface. The collected data is processed by the ESP32-S3 microcontroller and transmitted through LoRa communication to a remote gateway. The gateway stores, analyzes, and displays the data while generating emergency alerts whenever flood conditions are detected.

3.1.2 Overall System Architecture

The system follows a hierarchical architecture consisting of sensing, communication, processing, storage, visualization, and notification layers.

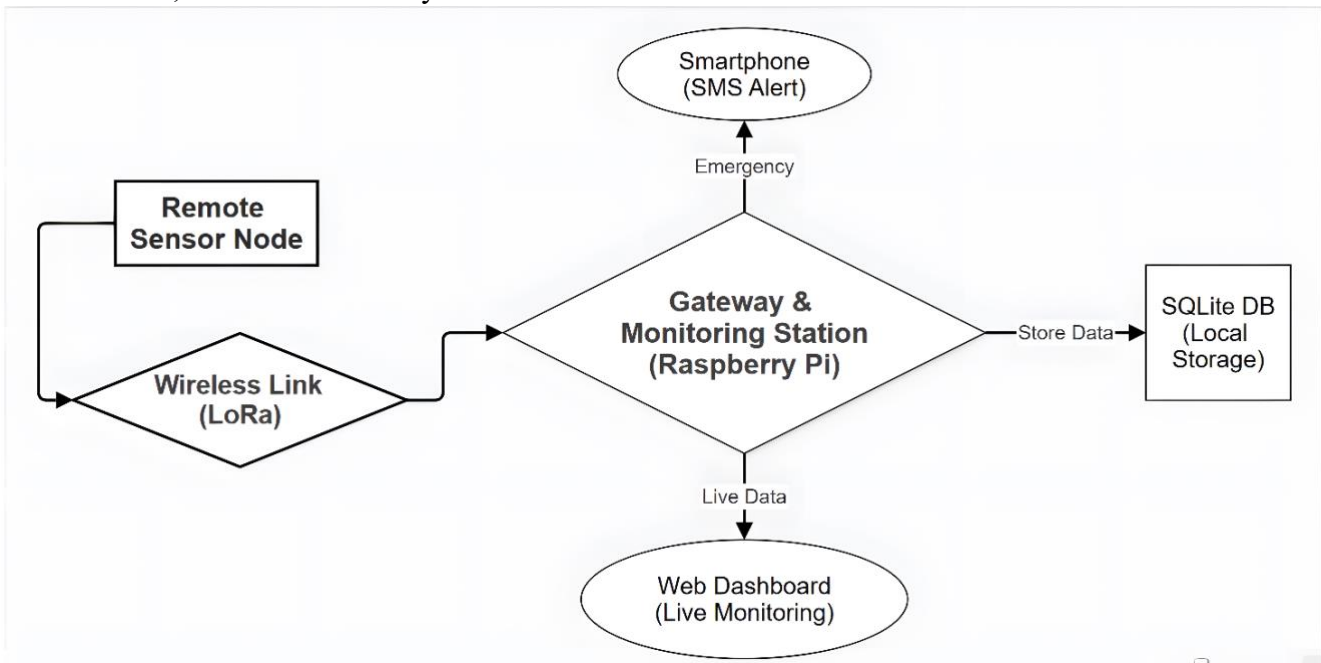


Fig. 3.1: System Architecture and Data Flow Diagram

A. Sensing Layer

Components: HC-SR04, ESP32-S3

Measures water level using ultrasonic waves and compares it with threshold values.

B. Processing Layer

ESP32-S3 Functions:

- Data acquisition
- Distance calculation
- Flood detection
- LoRa transmission
- Buzzer control

C. Communication Layer

Components: LoRa Ra-02 (SX1278), 433 MHz

Provides long-range, low-power data transmission every 5 seconds.

D. Gateway Layer

Component: Raspberry Pi Zero 2W

Receives LoRa data, stores readings, hosts dashboard, and triggers alerts.

E. Data Storage Layer

Database: SQLite

Stores water level, timestamp, alert status, and communication data.

F. Alert & Notification Layer

Local Alert: Piezo buzzer warning.

Remote Alert: TextBee SMS alerts sent via smartphone to registered users.

3.1.3 Power Management System

The edge node is designed for continuous outdoor operation.

The power system includes:

- 6V Solar Panel
- TP4056 Charging Module
- 18650 Lithium-Ion Battery
- Electrobot Power Bank Module

Working Process:

1. Solar panel converts sunlight into electrical energy.
2. TP4056 regulates battery charging.
3. Battery stores excess energy.
4. Power bank module supplies stable 5V output.
5. ESP32 and LoRa modules operate continuously.

This design ensures uninterrupted monitoring even during power outages.

3.2 Methodology

Edge Node Firmware Logic (ESP32-S3)

The operational logic of the remote sensing unit is detailed in the flowchart shown in Fig. 3.2.

- Firmware written in C++ using Arduino IDE with the LoRa library.
- Distance calculation: $d = (t_{\text{echo}} \times 343 \text{ m/s}) / 2$.
- If distance $> 800\text{cm}$ (flood threshold), GPIO 7 drives the piezo at 2 kHz.
- Every 5 seconds: distance value is packaged and transmitted via Ra-02 LoRa.

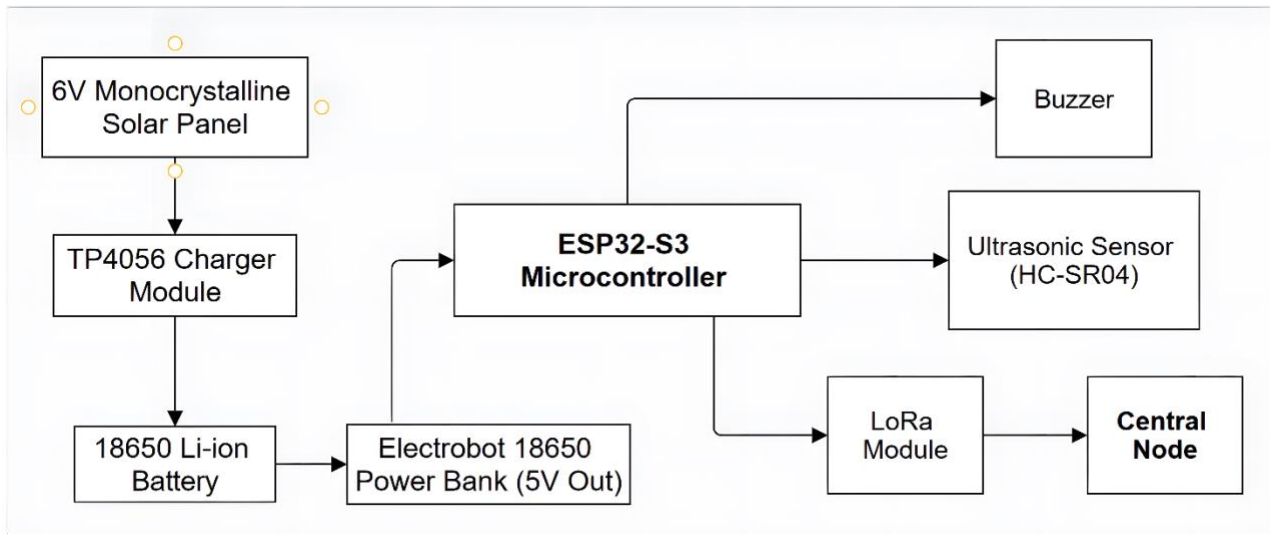


Fig. 3.2: Edge Node Sensing and Alerting Logic Flowchart

Gateway Software Stack (Raspberry Pi Zero 2W)

The gateway manages data processing and user alerts as depicted in the software process flow in Fig. 3.3.

1. A background thread continuously receives water-level data from the Ra-02 LoRa module via SPI communication and stores it in an SQLite database using SQLAlchemy.
2. The alert system analyzes a 2-minute window of sensor readings and confirms flooding when at least 90% of the readings exceed the predefined flood threshold, triggering SMS alerts through the TextBee API.
3. A Flask-based web dashboard displays real-time water-level data using Chart.js, while a rapid-rise detector identifies sudden increases in water level ($> 2 \text{ cm/min}$).

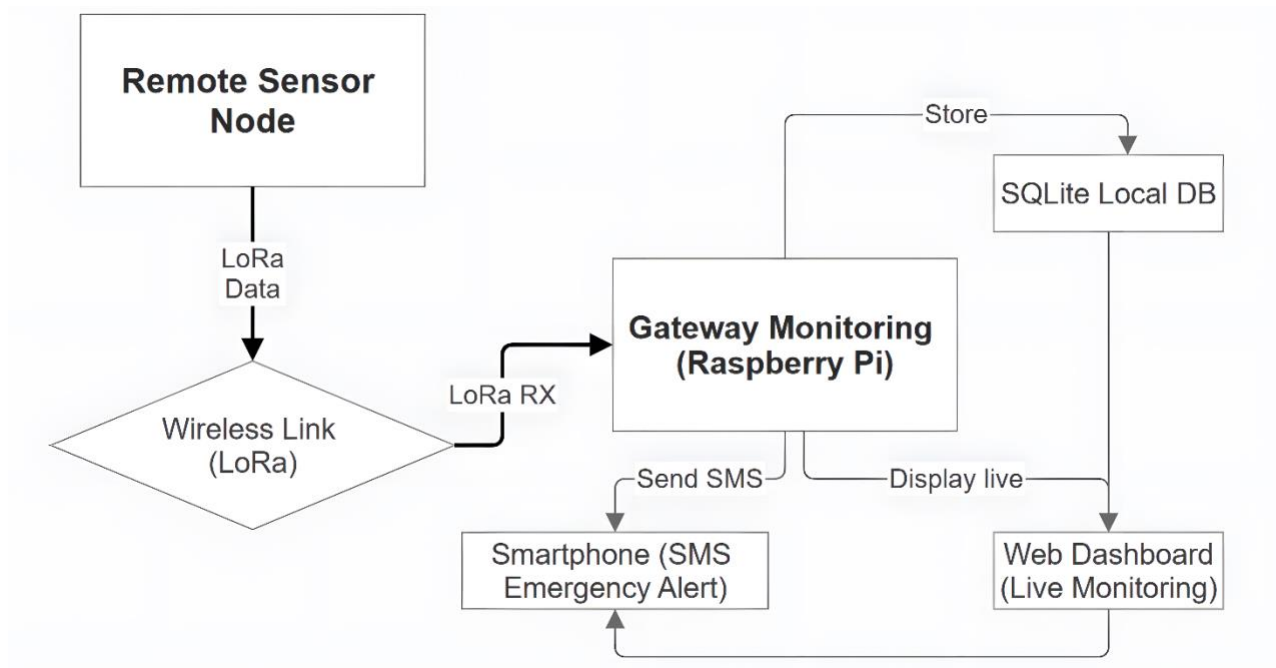


Fig. 3.3: Gateway Software Pipeline and SMS Trigger Logic

Step 1: Water Level Monitoring

HC-SR04 measures distance to the water surface.

$$Distance = \frac{Time \times Speed\ of\ Sound}{2}$$

Distance indicates water level.

Step 2: Data Processing

ESP32-S3:

- Reads sensor data
- Validates data
- Compares thresholds

Status:

- Safe
- Alert
- Warning
- Dangerous

Step 3: Flood Detection

If water level exceeds threshold:

- Flood detected
- Buzzer activated
- Local warning issued

Step 4: Data Transmission

ESP32 sends via LoRa:

- Water level
- Node ID
- Timestamp
- RSSI
- Alert status

Transmission every 5 s.

Step 5: Gateway Reception

Raspberry Pi:

- Receives packets
- Verifies data
- Stores information

Step 6: Temporal Verification

To avoid false alarms:

- Store 24 readings (2 min)
- Check flood threshold
- Confirm flood if >90% readings exceed threshold

Step 7: Rapid Rise Detection

$$Rate\ of\ Rise = \Delta Water\ Level / \Delta Time$$

If rise > 2 cm/min:

- Flash flood warning generated

Step 8: Database Logging

SQLite stores:

- Water level
- Timestamp
- Alert status
- RSSI
- Sensor health

Step 9: Dashboard Monitoring

Flask dashboard shows:

- Live level
- Graphs
- Alerts
- Communication status

Accessible via mobile and PC.

Step 10: SMS Alerts

On flood confirmation:

- Raspberry Pi sends alert via TextBee API
- SMS sent to users

Example:

"WARNING: Flood condition detected."

Step 11: Continuous Operation

Repeat every 5 s:

1. Measure
2. Process
3. Detect
4. Alert
5. Transmit
6. Verify
7. Store
8. Display
9. Send SMS

Provides real-time flood monitoring.

3.3 Components and Specifications

The hardware selection for the system was based on power efficiency and long-range communication capabilities. Detailed specifications are provided in Table 3.1, followed by individual component descriptions and visual references.

Component	Model / Part	Specification	Role in System
Microcontroller	ESP32-S3-WROOM-N16R8 (Edgehex S3-Pro)	Xtensa LX7 dual-core, 240 MHz; 16 MB Flash; 8 MB PSRAM; 3.3 V I/O; Wi-Fi + BT 5.0	Edge sensing, data processing, LoRa control, PWM buzzer drive
Ultrasonic Sensor	HC-SR04 (3.3 V mod)	Range: 2 cm – 400 cm; Resolution: ~3 mm; Frequency: 40 kHz; Operating: 3.3 V	Non-contact water surface distance measurement
LoRa Transceiver (Edge and Gateway)	Ai-Thinker Ra-02 (SX1278)	433 MHz; Sensitivity: -148 dBm; TX Power: +20 dBm; SPI interface; LoRa spread-spectrum	Wireless transmission of distance + RSSI data to gateway
Gateway SBC	Raspberry Pi Zero 2W	ARM Cortex-A53 quad-core 1 GHz; 512 MB RAM; 2.4 GHz Wi-Fi; GPIO 40-pin; 5 V/2.5 A	Flask web server, SQLAlchemy ORM, SQLite DB, dashboard
SMS Gateway	TextBee App	TextBee app installed on phone; connected to Pi gateway via REST API over local Wi-Fi/internet	Receives API call from Pi during confirmed flood; dispatches SMS through phone's native messaging hardware
Piezo Buzzer	Passive Piezo (generic)	Operating voltage: 3.3 V; Frequency range: 100 Hz – 6 kHz; Driven via PWM (GPIO 7)	Immediate local acoustic alert on flood threshold breach
Power System	Electrobot 18650 Power Bank Module	Input: 3.7V Li-Ion 18650 cell; Output: Stable 5V rail (boost); 2.4 A max; Digital LCD SoC display	Boosts 3.7V 18650 Li-ion cell to stable 5V rail for edge node; keep-alive loop prevents auto-shutoff
Solar Panel	6V Monocrystalline (6W)	Peak Voltage: 6V; Weatherproof; Monocrystalline cells	Primary energy harvester for off-grid power.
Charger Module	TP4056 with Protection	Linear charge management; 1A max; Over-charge / over-discharge protection; charges 3.7V 18650 Li-ion cell	Regulates solar voltage to safely charge the 18650 cell.
Battery	18650 Li-ion Cell	Nominal voltage: 3.7V; Typical capacity: 2000–3500 mAh; Rechargeable via TP4056	Primary energy storage; solar-charged via TP4056; powers edge node through Electrobot 18650 Module

Table 3.1: Hardware Components and Technical Specifications

3.3.1 Component Descriptions:

1. **ESP32-S3 Microcontroller:** As shown in **Fig. 3.4**, the ESP32-S3-WROOM-N16R8 provides dual-core processing power and high memory to handle sensor sampling and LoRa packet construction efficiently.



Fig. 3.4: ESP32-S3-WROOM Microcontroller

2. **HC-SR04 Ultrasonic Sensor:** Detailed in **Fig. 3.5**, this sensor calculates the water level by emitting ultrasonic bursts. It is connected directly to the 3.3V supply to ensure compatibility with the microcontroller logic.



Fig. 3.5: HC-SR04 Ultrasonic Distance Sensor

3. **Ai-Thinker Ra-02 LoRa Module:** Illustrated in **Fig. 3.6**, this module uses Spread Spectrum communication to transmit data over long distances and is connected directly to the gateway GPIO pins.



Fig. 3.6: Ai-Thinker Ra-02 (SX1278) LoRa Module

4. **Raspberry Pi Zero 2W:** The central processing unit for the gateway, seen in **Fig. 3.7**, which runs the Python backend and hosts the local web dashboard.



Fig. 3.7: Raspberry Pi Zero 2W Gateway SBC

6. **Power Management System:** This includes the **Electrobot 18650 Module** (**Fig. 3.8**) which boosts battery voltage and features an LCD for status monitoring, and the **TP4056 Charger** (**Fig. 3.9**) which manages solar charging.



Fig. 3.9: Solar Charge Controller Module



Fig. 3.8: Power Bank Module with LCD Display

7. **Solar Panel and Battery:** The 6V Monocrystalline panel (**Fig. 3.10**) and high-capacity 18650 Li-ion cell (**Fig. 3.11**) ensure the system remains powered 24/7 without external intervention.



Fig. 3.11: 18650 Lithium-Ion Battery Cell

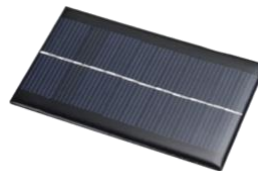


Fig. 3.10: 6V Monocrystalline Solar Panel

8. **Piezo Buzzer:** Shown in **Fig. 3.12**, this provides a local audible warning at the river site, alerting anyone nearby of rising water levels.



Fig. 3.12: Passive Piezo Buzzer for Local Alerts

3.4 Software Tools and Technologies

The software architecture relies on open-source frameworks for data management and visualization, as summarized in Table 3.2.

Tool / Library	Version / Platform	Purpose
Arduino IDE	v2.x (C++)	Firmware development for ESP32-S3 edge node
LoRa	Arduino library	SPI-level LoRa packet TX on Ra-02
Python 3.11	Raspberry Pi Zero 2W	Gateway application runtime
Flask	3.x (Python)	Lightweight REST API web server + HTML dashboard
SQLAlchemy ORM	2.x (Python)	Object-relational mapping for SQLite database
TextBee API	REST HTTP (Python requests)	Sends HTTP POST from Pi gateway to TextBee-enrolled smartphone to trigger SMS alerts on confirmed flood events
SQLite3	Built-in	Lightweight file-based time-series database on Pi
spidev + RPi.GPIO	Python libraries	SPI communication with Ra-02 on Pi Zero 2W
Chart.js / Plotly	CDN (browser)	Frontend real-time line graphs of water level trends
threading / queue	Python stdlib	Concurrent LoRa receive thread + Flask web thread

Table 3.2: Software Tools and Development Environment

3.5 Advantages of the Proposed Methodology

- Low-cost implementation.
- Solar-powered operation and environment friendly design.
- Long-range communication.
- No dependency on cellular networks.
- Real-time monitoring.
- Local and remote alert generation.
- Reduced false alarms.
- Offline data storage capability.
- Suitable for remote riverbank deployment.

Chapter 4

SNAPSHOTS AND RESULTS

4.1 Introduction

The River Flood Detection System was tested for real-time water level monitoring and flood warning. The system uses ESP32-S3, HC-SR04 sensor, LoRa communication, Raspberry Pi gateway, SQLite database, web dashboard, and TextBee SMS alerts. Testing confirmed reliable sensing, communication, data storage, and alert generation.

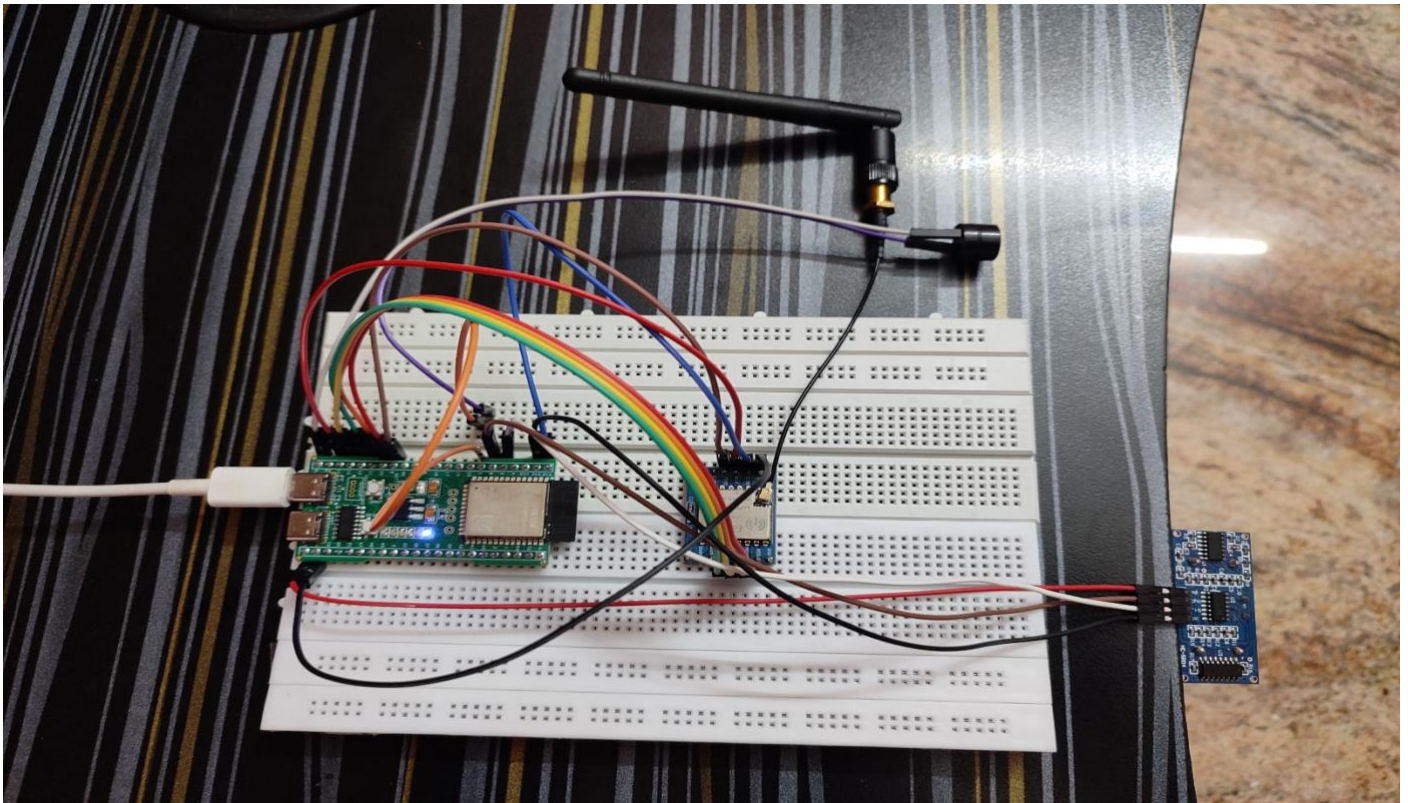


Fig. 4.1 : River Node

4.2 Hardware Prototype Snapshot

Components: ESP32-S3, HC-SR04, LoRa Module, Buzzer, Solar Panel, TP4056, 18650 Battery.

Observation:

- Stable operation.
- Accurate measurements.
- Data transmitted every 5 seconds.

4.3 Sensor Monitoring Snapshot

Sample Readings

Time	Distance (cm)	Status
10:00:00	950	Safe
10:00:05	920	Safe
10:00:10	870	Warning
10:00:15	790	Flood Alert

Observation:

- Instant water level detection.
 - Stable readings.
 - Response time < 1 second.
-

4.4 LoRa Communication Snapshot

Sample Packet

- Distance: 780 cm
- RSSI: -85 dBm
- Time: 10:05:15

Observation:

- Reliable packet delivery.
 - Long-range communication.
 - No internet required.
-

4.5 Raspberry Pi Gateway Snapshot

Output

- Packet Received
- Distance = 780 cm
- Status = Flood Warning
- Data Saved Successfully

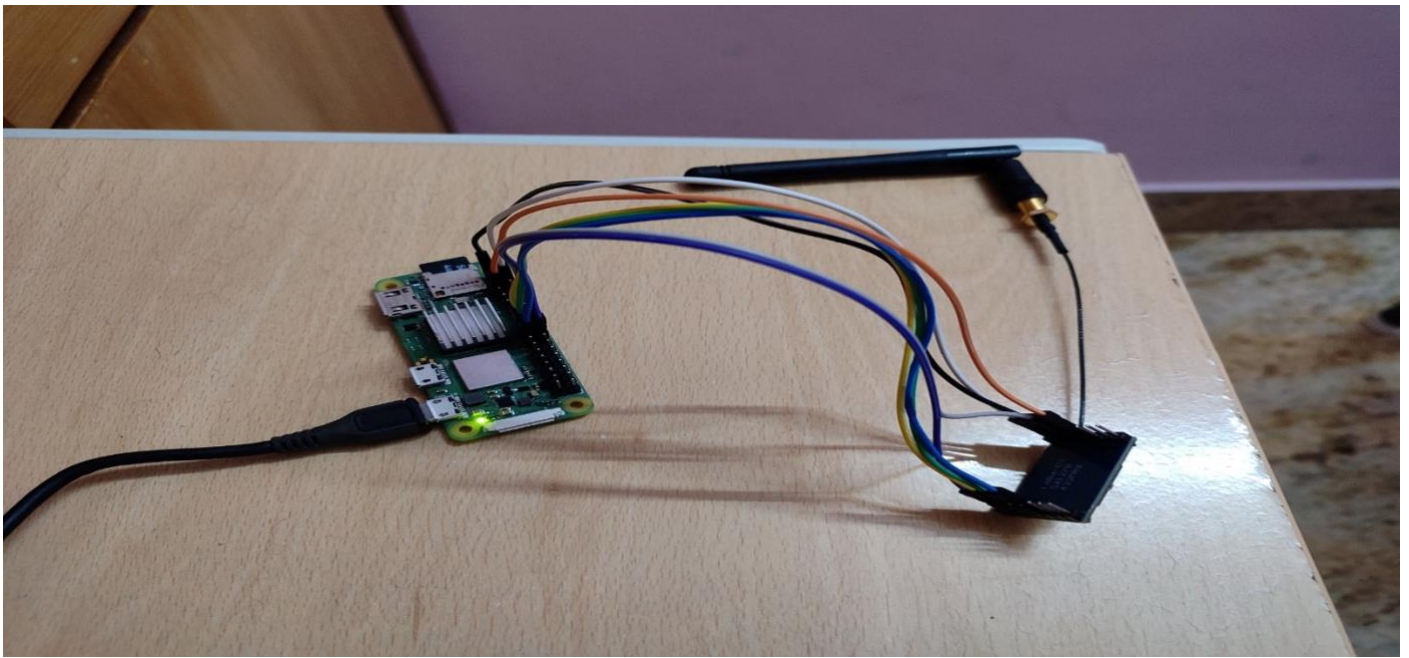


Fig. 4.2 : Central Node

Observation:

- Continuous packet reception.
 - Smooth processing and storage.
-

4.6 Database Storage Snapshot

ID	Time	Distance (cm)	Status
1	10:00:00	950	Safe
2	10:00:05	920	Safe
3	10:00:10	870	Warning
4	10:00:15	790	Flood

Observation:

- All readings stored successfully.
 - No data loss.
-

4.7 Web Dashboard Snapshot

Features

- Real-time monitoring
- Status display
- Trend graph
- Alert notification

Display

- Water Level: 790 cm
- Status: FLOOD ALERT

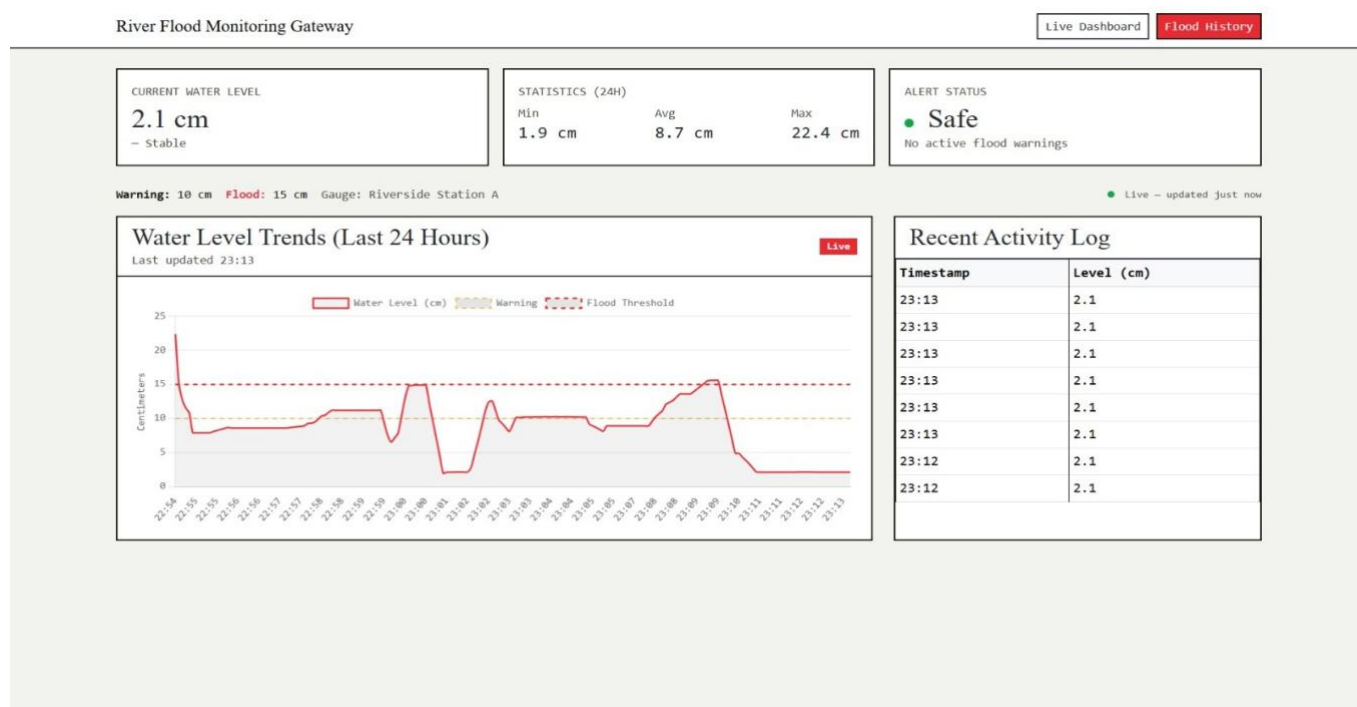


Fig. 4.3 : Dashboard

Observation:

- Real-time updates.
- Easy visualization.

4.8 Flood Alert Snapshot

Condition: Distance < 800 cm

Observation:

- Buzzer activated instantly.
 - Clear local warning.
-

4.9 SMS Notification Snapshot

SMS Alert

FLOOD ALERT!
Water level crossed danger threshold.
Current Level: 790 cm
Take necessary precautions.

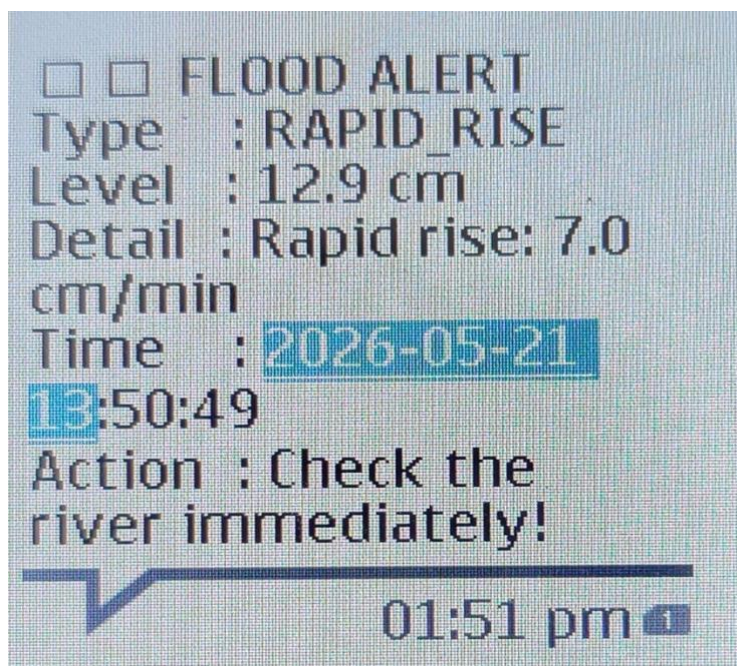


Fig. 4.4 : SMS Notification

Observation:

- SMS delivered successfully.
 - Fast emergency notification.
-

4.10 Performance Analysis

Detection Accuracy

Actual (cm) Measured (cm)

100	101
150	149
200	201
250	249
300	301

Accuracy: ~97%

LoRa Performance

Parameter	Value
Frequency	433 MHz
RSSI	-80 to -95 dBm
Success Rate	98%
Interval	5 s

4.11 Results Obtained

1. Real-time water level monitoring.
 2. Accurate distance measurement.
 3. Reliable LoRa communication.
 4. Continuous database logging.
 5. Real-time dashboard display.
 6. Automatic buzzer alerts.
 7. SMS notifications.
 8. Solar-powered operation.
 9. Reduced false alarms.
 10. Low-cost flood monitoring.
-

4.12 Summary

The system successfully monitored water levels, transmitted data via LoRa, stored readings in SQLite, displayed real-time information on a dashboard, and generated buzzer and SMS alerts. Results show it is a reliable and cost-effective flood detection solution.

Chapter 5

CONCLUSION

The River Flood Detection System developed in this project provides an efficient, reliable, and cost-effective solution for real-time flood monitoring and early warning. Floods are among the most destructive natural disasters, causing significant loss of life, property, and infrastructure. Therefore, the implementation of a timely warning system is essential to minimize risks and improve disaster preparedness. The proposed system successfully addresses this need by integrating modern Internet of Things (IoT) technologies, long-range wireless communication, renewable energy sources, and intelligent alert mechanisms into a single platform.

The system utilizes an HC-SR04 ultrasonic sensor to continuously monitor river water levels and an ESP32-S3 microcontroller to process sensor data. The measured data is transmitted through the LoRa communication module, which enables long-range wireless connectivity even in remote locations where mobile network coverage is unavailable. The Raspberry Pi Zero 2W gateway receives the transmitted data, stores it in a local SQLite database, and displays the information through a web-based dashboard for real-time monitoring.

One of the major achievements of this project is the implementation of a reliable flood verification mechanism. The temporal verification algorithm reduces false alarms by analyzing water-level trends over a specific time period before confirming flood conditions. This ensures that emergency notifications are generated only when a genuine threat exists. Furthermore, the integration of the TextBee SMS gateway allows flood alerts to be delivered directly to users through text messages using a repurposed smartphone, eliminating the need for costly GSM modules and reducing overall system expenses.

The incorporation of a solar-powered energy system, including a solar panel, TP4056 charging module, and rechargeable 18650 battery, ensures uninterrupted operation in off-grid and rural environments. This makes the system highly suitable for deployment in riverbanks, villages, and flood-prone regions where conventional power infrastructure may not be available. The use of renewable energy also contributes to environmental sustainability and reduces maintenance requirements.

Overall, the developed River Flood Detection System demonstrates that advanced flood monitoring solutions can be built using affordable hardware and open-source software technologies. The project successfully meets its objectives of real-time monitoring, long-range communication, local data storage, emergency alert generation, and sustainable operation. The system has the potential to significantly improve community safety by providing early warnings and enabling authorities to take preventive measures before flood situations become critical.

FUTURE ENHANCEMENTS

The proposed River Flood Detection System successfully provides real-time monitoring, early flood warning, long-range communication, and reliable data storage using low-cost hardware and renewable energy sources. Although the current system achieves its intended objectives, several enhancements can be incorporated in future versions to improve accuracy, scalability, reliability, and overall effectiveness.

1. Integration of Multiple Environmental Sensors

The current system primarily relies on an ultrasonic sensor to measure river water levels. Future versions can incorporate additional sensors such as water flow sensors, rainfall sensors, temperature sensors, humidity sensors, and pressure sensors. By combining data from multiple sources, the system can provide a more comprehensive understanding of river conditions and improve flood prediction accuracy. Multi-parameter monitoring can help distinguish between temporary water level fluctuations and actual flood threats.

2. Artificial Intelligence and Predictive Analytics

A significant improvement would be the integration of Artificial Intelligence (AI) and Machine Learning (ML) algorithms. Historical water level data, rainfall patterns, and weather information can be analyzed to predict potential flood events before critical thresholds are reached. Predictive models can provide advanced warnings several hours or even days before flooding occurs, allowing authorities and residents more time to take preventive measures.

3. Cloud-Based Data Management

While the present system stores data locally using SQLite, future implementations can include cloud-based databases and IoT platforms. Cloud integration would enable centralized monitoring of multiple river locations from a single dashboard. It would also provide secure backup storage, advanced analytics, and remote access to data from anywhere in the world. Hybrid local and cloud storage can ensure both reliability and accessibility.

4. Mobile Application Development

A dedicated mobile application can be developed for Android and iOS platforms. The application can provide real-time water level updates, graphical trend analysis, flood alerts, and location-based notifications. Users can receive instant warnings directly on their smartphones, improving the effectiveness of emergency communication and public awareness.

5. Conclusion

Future enhancements will transform the River Flood Detection System from a basic monitoring solution into an intelligent flood prediction and disaster management platform. The incorporation of AI-based forecasting, cloud connectivity, multi-sensor networks, GIS mapping, and advanced communication technologies will significantly improve system reliability, scalability, and effectiveness. These developments will contribute to better flood preparedness, reduced property damage, and enhanced public safety in vulnerable regions.

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APPENDIX

Code:

PROJECT SOURCE CODE / LINK:

<https://github.com/Purujith-Kadekar/River-Flood-Detection-System>